

1.1)

The basic tasks that all software engineering projects must handle are:

- Requirements Gathering
- High-Level Design
- Low-Level Design
- Development
- Testing
- Deployment
- Maintenance
- Wrap-up

1.2)

- Requirements Gathering: Learn what the customer wants and needs.
- High-Level Design: Description of the major pieces of the application and how they interact.
- Low-Level Design: Detailed description of how to build the pieces of the application so that the programmers can actually implement them.
- Development: Writing the code to implement the application.
- Testing: Verifying that the application functions properly under various conditions to identify any potential flaws or bugs.
- Deployment: Release the application to the users.
- Maintenance: Post-launch work on the project, such as bug fixes, additions, enhancements, and future versions of the program.
- Wrap-up: Evaluate what went right and what went wrong in the project so that you can repeat the good things and avoid the bad things in future projects.

2.4)

Google Docs also has an edit tracking tools that let you see who has modified a document and when. Changes are tracked automatically on Google Docs, where as gitHub requires intentional pushes to log changes made to the code. Both tools let you roll back documents to previous versions.

2.5)

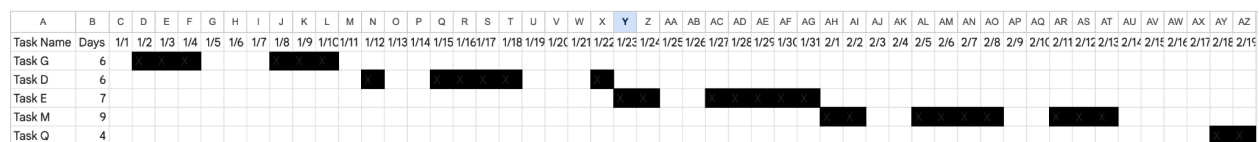
JBGE stands for Just Barely Good Enough. It refers to a system of writing code documentation that satisfies the requirements but doesn't have any additional/unnecessary features.

4.2)

- The critical path has a total expected length of 32 working days.
- The critical path is $G \rightarrow D \rightarrow E \rightarrow M \rightarrow Q$.
- The total expected length is 32 working days (same as the critical path).

4.4)

Gantt chart for critical path only



4.6)

You can treat these problems by adding extra tasks at the end of the schedule (or throughout it) to account for unexpected problems. If/when one of these problems does occur, insert its lost time into the schedule as part of those extra tasks.

4.8)

Rewrite this:

The biggest mistake you can make while tracking tasks is not taking action when a task slips. At a minimum, you need to pay closer attention to the task so that you can take action if it's in trouble.

The second biggest mistake is piling more people on the task and assuming they can cut the total time. Unless the new people have particularly useful expertise, bringing them up to speed may make the task take even longer.

5.1)

Five characteristics of good requirements are: clear, unambiguous, consistent, prioritized, and verifiable.

5.3)

Audience categories (B = Business, U = User, F = Functional, N = Nonfunctional, and I = Implementation)

- Allow users to monitor uploads/downloads while away from the office. (B)

- b. Let the user specify website log-in parameters such as an Internet address, a port, a username, and a password. (U, F)
- c. Let the user specify upload/download parameters, such as the number of retries if there's a problem. (U, F)
- d. Let the user select an Internet location, a local file, and a time to perform the upload/download. (U, F)
- e. Let the user schedule uploads/downloads at any time. (N)
- f. Allow uploads/downloads to run at any time. (N)
- g. Make uploads/downloads transfer at least 8 Mbps. (N)
- h. Run uploads/downloads sequentially. Two cannot run at the same time. (N)
- i. If an upload/download is scheduled for a time when another is in progress, the new task waits until the other one finishes. (N)
- j. Perform scheduled uploads/downloads. (F)
- k. Keep a log of all attempted uploads/downloads and whether they succeeded. (F)
- l. Let the user empty the log. (U, F)
- m. Display reports of upload/download attempts. (U, F)
- n. Let the user view the log reports on a remote device, such as a phone. (U, F)
- o. Send an email to an administrator if an upload/download fails more than its maximum retry number of times. (U, F)
- p. Send a text message to an administrator if an upload/download fails more than its maximum retry number of times. (U, F)

All the categories include at least one requirement except for the implementation requirements category, which is empty since the project would likely have finished implementation by the time these features are being used.

5.9)

Must

- Sound effects when buttons are pressed and when the game is won/lost

Should

- Some form of score keeping to track how many games you have won/how many guesses it took to win each game on average

Could

- A hint button that automatically picks one letter that is in the word

Won't

- A 3D render of Mr. Bones that does a fully choreographed dance on screen when the game ends